OpsPilot: Product Requirements

1. Problem and target users

A growing retail company runs its operations on email, spreadsheets and personal follow-ups. A request such as “500 display units for the Pune launch will be 10 days late, approve the extra cost by 4 PM” has to be read, checked against stock, assessed, costed, approved and reported by hand. This is slow, depends on who happens to be online, and leaves no reliable record of who decided what.

User	Need
Operations analyst	Turn an email into checked facts and options quickly; chase missing information
Manager, regional head	See only what needs their decision, with the evidence and the cost, and decide within their limit
Admin	Control the system, test failure paths, see the audit trail
Senior leadership	A short summary of open cases, risks and overdue approvals

2. Proposed workflow

1. **Receive.** An email is imported from the inbox, or a request is typed in.
2. **Extract.** The intake agent returns the key facts (place, item, quantities, supplier, delay, deadline, ask) and an exact quote for each.
3. **Verify.** Code checks required fields, whether each quote is really in the email, inventory figures, duplicates and prompt-injection patterns.
4. **Assess.** The risk agent rates urgency and impact; a rule-based score is computed beside it; the higher level wins.
5. **Recommend.** The planner proposes two or three options with extra cost, timing, rationale and risks.
6. **Review.** A person accepts the recommendation, picks another option or writes their own. Overrides need a reason.
7. **Approve.** A cost above zero, a high or critical risk, or low confidence creates an approval for the lowest role whose limit covers the cost. The person who chose the action cannot approve it. Overdue approvals escalate to the next role.
8. **Execute and close.** Work starts once the action is chosen and approved; then resolve and close. Draft emails for the requester, supplier, store team or approver are written by the communications agent; a person sends them.
9. **Report.** The reporting agent summarises one case or the whole portfolio.

Every step is written to an append-only audit trail.

3. Requirements

Functional

#	Requirement
F1	Read a request from a sample inbox or a manual form
F2	Extract facts, dates and dependencies, each with supporting evidence
F3	Assess urgency, impact and risk; show agent level, rule level and final level
F4	Flag missing, conflicting, duplicate, unsupported and unsafe content

#	Requirement
F5	Recommend options with cost and rationale; allow review and override
F6	Route approvals by amount; escalate when overdue; reject with reason and re-plan
F7	Track status: processing, needs info, review, pending approval, in progress, resolved, closed, cancelled
F8	Draft stakeholder emails and management summaries
F9	Role-based login; analyst cannot approve; limits Rs 50,000 / Rs 5,00,000 / unlimited
F10	Audit trail of agent runs and human actions
F11	Admin tools: advance the clock, simulate an AI outage

Non-functional

Area	Requirement
Safety	Agents never approve or send; email text is treated as untrusted data; model output is validated before use
Auditability	Audit rows are never updated or deleted; agent runs record model, duration, attempt and prompt version
Security	Hashed passwords, session cookie (HttpOnly, SameSite), server-side role checks, parameterised SQL, escaped output in the UI
Availability	If the AI service fails, requests stay usable: the user can retry or fill fields by hand
Performance	Analysis in under a minute per request; the interface stays responsive while it runs
Operability	One process, one SQLite file, health endpoint, container image

4. Assumptions and trade-offs

Decision	Reason	Cost
Own small pipeline instead of an agent framework	Five agents with fixed order; easier to explain, test and audit	No built-in tooling for dynamic planning
Code decides what the model must not (grounding, limits, final risk, confidence)	Model output can be wrong or manipulated	More rules to maintain
SQLite and a single process	Fast to run and demo	Not for many concurrent users
Model reached through a local command line tool, one call at a time	No API key handling; small server	Slow under load
Mock inbox and outbox	The brief does not require real mail	Real mail needs IMAP/SMTP and retries
Costs are model estimates	No price list in scope	Approvers must check the figures
Indian rupees, IST	Matches the scenario	No multi-currency

5. Edge-case handling

All fifteen listed scenarios are handled. Mail sending is a mock with a switch that makes it fail, to show the failure path.

Scenario	Detection	Response
Missing information	Required-field check	Flag, status “needs info”, draft email asking for the fields
Conflicting information	Email quantity vs inventory	Flag with both numbers; confidence lowered
Duplicate request	Same place and item, or similar text	Flag linking the earlier request
Unclear intent	Several intents, no clear ask	All intents listed; missing fields flagged
Incorrect extraction	User edits a field	Edit logged with reason; analysis re-runs
Low-confidence recommendation	Confidence below 0.6	No preselected option; choice needs a reason and an approver
Approval delay	Due time passes	Escalation to next role; new approval; audit entry
Approval rejection	Approver rejects with reason	Option marked rejected; planner proposes alternatives
Changing circumstances	Supplier update	Pending approval cancelled; analysis re-run
Unavailable AI service	Call error, timeout, bad output twice	Flag; request stays open; retry
Unauthorized action	Role limit check; approving own choice	Refused with 403; attempt in audit
Communication failure	Mail server error	Message marked failed; retry send; audit entry
Prompt injection	Patterns and model flag	Flag; instructions ignored; human review
Request cancellation	Cancel button	All approvals cancelled; approver notified by draft
Contradictory agents	Risk levels two or more apart	Flag; higher level used

6. Success metrics and MVP scope

Targets for a pilot (not measured results):

Metric	Target
Time from receipt to a reviewed recommendation	Under 10 minutes, from about an hour by hand
Extracted fields corrected by users	Under 10%
Recommendations accepted without override	Above 60%
Approvals escalated for delay	Falling month on month
Requests with a complete audit trail	100%
Unauthorized approvals completed	0

In the MVP: everything in section 3, the six sample emails, the demo roles, the admin tools, tests and a container image.

Out of the MVP: live mailbox and real outbound email, single sign-on, session expiry, price and supplier catalogues, multi-currency, mobile app, notifications by chat, analytics dashboards.